

물리화학 Ⅲ (이.공) (CHEM304-01)

학기	2026년 2학기
담당교수	김종구
이수구분	교선평수
학점	3.0
강의시간	화13:00-14:30(E7 - G01), 목13:00-14:30(E7 - G01)
강의실	E2 608 (교수실)
소속	기초학부

교과개요

The objective of the Physical Chemistry 3 course is to extend the fundamental concepts introduced in Physical Chemistry 1 and 2 and to deepen students' understanding of the behavior of molecules in various states: solids, liquids, and gases. In particular, students will learn essential concepts of reaction kinetics and reaction dynamics, which are crucial for understanding a wide range of chemical and biological catalytic reactions. Students will also explore practical experimental and theoretical methods used to investigate reaction mechanisms. Because this course builds substantially on the material covered in Physical Chemistry 1 and 2, prior completion of these courses is strongly recommended.

교과목표

This course covers a broad range of topics related to molecular interactions and dynamics. These include the fundamental principles of intermolecular interactions, the physical properties of soft matter and solid-state materials, molecular motion, chemical kinetics, molecular reaction dynamics, and surface processes.

주차별 강의계획

1주차:Molecular interactions - 1 (To be replaced with a recorded video lecture)
2주차:Molecular interactions - 2 (To be replaced with a recorded video lecture)
3주차:Molecular interactions - 3 / Solids - 1
4주차:Solids - 2
5주차:Solids - 3 / Molecules in motions - 1
6주차:Molecules in motions - 2
7주차:Molecules in motions - 3
8주차:Mid-term exam
9주차:Chemical kinetics - 1
10주차:Chemical kinetics - 2
11주차:Chemical kinetics - 3
12주차:Chemical kinetics - 4
13주차:Reaction Dynamics - 1
14주차:Reaction Dynamics - 2
15주차:Reaction Dynamics - 3
16주차:Final exam

학습윤리

Cheating: Students who cheat in exams cannot pass this course.

교과정책(방침)

More than 8 classes absent = failure. 3 times late = 1 absence.

Your absence is excused only when you missed class because of these three reason: One, you were involved in a university-sponsored activity; Two, there was a death in your family; or Three, you were sick and stayed overnight at a hospital. For each of these case, ou need to bring me official letters.